

FREEDOM INTERNATIONAL SCHOOL

WORKSHEET- SOLUTIONS

PHYSICS

CLASS XII

MAGNETISM AND MATTER

1. Two unlike poles of strength 10 Am each are held in air at a distance of 0.10 m from each other. The magnetic force of attraction between them is

(a) 3×10^{-3} N (b) 2.5×10^{-3} N (c) 1.5×10^{-3} N (d) 1×10^{-3} N

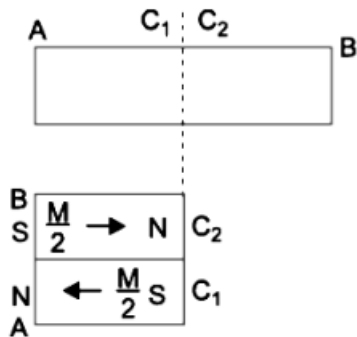
Ans: (d) 1×10^{-3} N

$$F = \frac{\mu_0}{4\pi} \frac{q_{m1} q_{m2}}{r^2}$$

2. A bar magnet AB of magnetic moment $\vec{m}$ is cut into two equal parts perpendicular to its axis. One part is kept over the other so that end B is exactly over A. The magnetic moment of the combination so formed will be

(a) zero (b) $\vec{m}$ (c) $\vec{m}/4$ (d) $3\vec{m}/4$

(a) zero



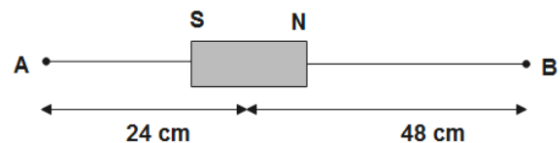
The magnetic moment of each half bar magnet is $\frac{M}{2}$ but oppositely directed, so

$$\text{net magnetic moment of combination} = \frac{M}{2} - \frac{M}{2} = 0 \text{ (zero).}$$

3. A bar magnet of length 3 cm has points A and B along its axis at distances of 24 cm and 48 cm on the opposite sides. Ratio of magnetic fields at these points will be

(a) 8 (b) $1/2\sqrt{2}$

Ans : (a) 8



(c) 3 (d) 4

For a short magnet at a point along its axial line, magnetic field is given by

$$B = \frac{\mu_0}{4\pi} \cdot \frac{2M}{r^3}$$

$$\Rightarrow \frac{B_1}{B_2} = \left(\frac{r_2}{r_1}\right)^3 = \left(\frac{48}{24}\right)^3 = \frac{8}{1} = 8$$

4. The couple acting on a magnet of length 10 cm and pole strength 15 Am, kept in a field of $B = 2 \times 10^{-5}$ T, at an angle of 30° is

(a) 1.5×10^{-5} Nm (b) 1.5×10^{-3} Nm (c) 1.5×10^{-2} Nm (d) 1.5×10^{-6} Nm

Ans: (a) 1.5×10^{-5} Nm

$$\text{Torque} = mB \sin\theta = (q_m 2l)B \sin\theta$$

5. Two wires of same length are shaped into a square and a circle. If they carry same current, ratio of the magnetic moments is

(a) $2: \pi$ (b) $\pi: 2$ (c) $\pi: 4$ (d) $4: \pi$

Ans: (c) $\pi: 4$

Suppose length of each wire is l .

$$A_{\text{square}} = \left(\frac{l}{4}\right)^2 = \frac{l^2}{16}$$

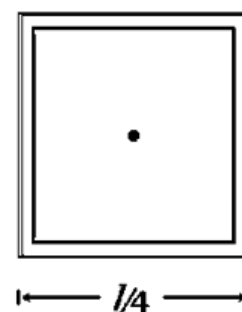
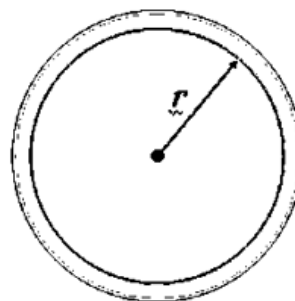
$$A_{\text{circle}} = \pi r^2 = \pi \left(\frac{l}{2\pi}\right)^2 = \frac{l^2}{4\pi}$$

$\therefore$ Magnetic moment

$$M = iA$$

$$\Rightarrow \frac{M_{\text{square}}}{M_{\text{circle}}} = \frac{A_{\text{square}}}{A_{\text{circle}}}$$

$$\frac{l^2/16}{l^2/4\pi} = \frac{\pi}{4}$$



6. Which of the following is weakly repelled by a magnetic field?

(a) Iron (b) Steel (c) Cobalt (d) Copper

Ans: (d) Copper

7. Magnetic permeability is maximum for

(a) ferromagnetic substances (b) diamagnetic substance
(c) paramagnetic substances (d) all of these

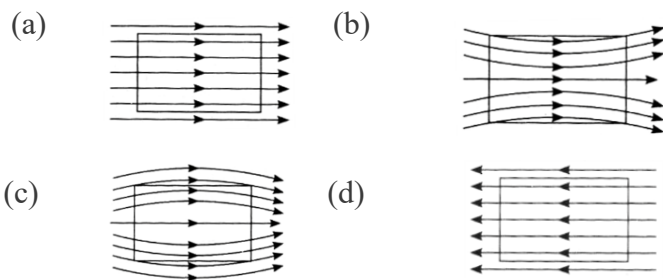
Ans: (a) ferromagnetic substances

8. Susceptibility of ferromagnetic substance is

(a) >1 (b) <1 (c) zero (d) 1

Ans: (a) >1

9. A uniform magnetic field parallel to the plane of paper, existed in space initially directed from left to right. When a bar of soft iron is placed in the field parallel to it, the lines of force passing through it will be represented by figure



Ans: (b)

10. A bar magnet of magnetic moment 10^4 J/T is free to rotate in a horizontal plane. The work done in rotating the magnet slowly from a direction parallel to a horizontal magnetic field of $4 \times 10^{-5} \text{ T}$ to a direction of 60° from the field will be

(a) 0.2 J (b) 2 J (c) 4.18 J (d) $2 \times 10^2 \text{ J}$

Ans: (a) 0.2 J

$$\theta_i = 0^\circ \text{ to } \theta_f = \theta$$

Work done is given by,

$$W = \mathbf{mB}(1 - \cos \theta)$$

$$\text{So, } W = 10^4 \times 4 \times 10^{-5} (1 - \cos 60^\circ)$$

$$W = 2 \times 10^{-1} = 0.2 \text{ J}$$

11. A paramagnetic substance of susceptibility 3×10^{-4} is placed in a magnetizing field of $4 \times 10^{-4} \text{ A/m}$. Then the intensity of magnetization in the units of A/m is

(a) 1.33×10^8 (b) 0.75×10^{-8} (c) 12×10^{-8} (d) 14×10^{-8}

Ans: (c) 12×10^{-8}

$$\begin{aligned} \mathbf{M} &= \chi H = (4 \times 10^{-4}) \times (3 \times 10^{-4}) \\ &= 12 \times 10^{-8} \text{ A m}^{-1} \end{aligned}$$

12. If the magnetizing field on a ferromagnetic material is increased, its permeability

(a) is decreased (b) is increased (c) is unaffected (d) may be increased or decreased

Ans: (a) is decreased

Magnetic permeability of a ferromagnetic substance, $\mu = B/H$ or $\mu \propto \frac{1}{H}$, where H is the magnetising field. If H increases then μ decreases.

13. For a paramagnetic material, the dependence of the magnetic susceptibility χ on the absolute temperature is given as

(a) $\chi \propto T$ (b) $\chi \propto 1/T^2$ (c) $\chi \propto 1/T$ (d) independent

Ans: (c) $\chi \propto 1/T$

For questions 14 to 20, two statements are given- one labelled Assertion (A) and the other labelled Reason (R). Select the correct answer to these questions from the options as given below.

- A. Both Assertion and Reason are true and Reason is the correct explanation of Assertion.**
- B. Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.**
- C. Assertion is true but Reason is false.**
- D. Both Assertion and Reason are false.**

14. **Assertion:** Magnetic susceptibility is a pure number.

Reason: The value of magnetic susceptibility for vacuum is one.

Ans: C

15. **Assertion:** When radius of a circular loop carrying current is doubled, its magnetic moment becomes four times.

Reason: Magnetic moment depends on the area of the loop.

Ans: A

16. **Assertion:** The poles of magnet cannot be separated by breaking into two pieces.

Reason: The magnetic moment will be reduced to half when a magnet is broken into two equal pieces.

Ans: B

17. **Assertion:** The basic difference between an electric line and magnetic line of force is that former is discontinuous and the latter is continuous or endless.

Reason: No electric lines of forces exist inside a charged body but magnetic lines do exist inside a magnet.

Ans: A

18. **Assertion:** Magnetic moment of an atom is due to both, the orbital motion and spin motion of every electron.

Reason: A charged particle at rest produces a magnetic field.

Ans: C

19. **Assertion:** A paramagnetic sample displays greater magnetization (for the same magnetizing field) when cooled.

Reason: The magnetization does not depend on temperature.

Ans: C

20. **Assertion:** Above the Curie temperature, a ferromagnetic material becomes paramagnetic.

Reason: When a magnetic material is heated to very high temperature, it loses its magnetic properties.

Ans: A